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Probability Iterative Closest Point Algorithm for Position Estimation+

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Abstract— **This paper proposes probability iterative closest point (ICP) method based on expectation maximization (EM) estimation for point set registration with noise. The classical ICP algorithm can deal with rigid registration between two point sets effectively, but always fails to register point sets with noise. In order to improve the registration precision, a Gaussian model is introduced into the rigid registration. In each iteration, the classical ICP algorithm includes two steps, building the corresponding relationship and computing the rigid transformation. Similar to the traditional ICP, at each step, firstly the corresponding relationship is set up. Secondly, the rigid transformation is solved by singular value decomposition (SVD) method, and then the Gaussian model is updated by the distance and variance between two point sets. The experimental results on part B of CE-Shape-1 database and real position dataset validate that the proposed algorithm is more accurate.**

I. INTRODUCTION

Point set registration has become a very important and basic research subject in computer vision and image analysis due to its wide applications, such as simultaneous localization and mapping, 3D reconstruction, and medical image analysis. The typical method is the iterative closest point (ICP) algorithm which offers a good solution to the general point set registration [1], [2]. The ICP algorithm does not need to extract point set features, and also does not need data preprocessing. The ICP algorithm has been widely used in many fields for its advantages of high speed and precision.

In the past few decades, scholars made a lot of explorations for rigid point set registration. To speed up the ICP algorithm, Xu et al. [3] introduced five constraint conditions of registration point pairs, and Kim et al. [4] proposed two acceleration techniques: hierarchical model point selection and logarithmic data point search. Meanwhile, there were also many researchers studying the robustness of ICP.

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Almhdie et al. [5] presented an enhanced implementation of the popular ICP algorithm for the registration of 3D free-form closed surfaces, which was based on the use of a look up matrix for finding the best correspondence pairs. Zhang et al. [6] presented a more robust ICP approach for 2D point set registration and an inequality constraint of the rotation angle was introduced into the least square registration model which was solved by an extended ICP algorithm.

It should be pointed out that the above mentioned approaches have not handled the point sets with outliers and noise which exist widely in application. To solve the registration of point sets with outliers, many researchers made great efforts. Some matched corresponding points based on overlapping percentage. A new distance measure, fractional root mean squared distance which simultaneously computed the best transformation and the overlapping percentage could identify and discard outliers [7]. However, this method depended greatly on a parameter. Thus, Du et al. [8] proposed a novel objective function which could automatically compute rigid transformation, correspondence, and overlapping percentage without influence of a parameter. Meanwhile, the distance threshold was also discussed. A modified ICP method was proposed to ameliorate the performance of laser scan alignment by applying dynamic distance error threshold [9]. Ridene and Goulette proposed a variant of ICP based on an adaptive dynamic threshold and a RANSAC removing outlier process [10].

Due to the precision of the acquisition equipment, the acquired data may have different precision and random noise in the real application. The above-mentioned approaches are effective for outliers, but they are not suitable for dealing with point sets with noise. Therefore, some researchers proposed to join probability into the ICP algorithm to deal with it. A new method is corresponded to ICP with multiple matches weighted by normalized Gaussian weights, giving birth to the EM-ICP using expectation maximization (EM) principles [11], which adopts full correspondence relationship for all the points in the model point set and the shape point set. Hence, it is time-consuming and the efficiency is very low. To cope with this problem, we use the EM principle and adopt one-to-one correspondence which is for all the points in the shape point set only needed to find the closest points from the model set. This method achieves high efficiency. To prevent the proposed algorithm from being trapped into the local minimum, the variance of Gaussian probability model is updated from large to small step by step, which can make the registration from global to local.

This paper is organized as follows. Section II briefly reviews the process of the original ICP. Following that is Sec-

tion III, aiming at solving the rigid registration of point sets with noise, the Gaussian probability model is introduced and the probability iterative closest point algorithm is proposed. Experimental results on part B of CE-Shape-1 database and real position dataset are present in Section IV. Finally, the conclusion is given in Section V.

II. ITERATIVE CLOSEST POINT ALGORITHM

For the rigid registration of point sets, ICP is the typical algorithm proposed by Besl and McKay [1]. Moreover, the ICP algorithm has been widely used in various research fields for its fast speed and high accuracy, which will be introduced as follows.

Given two point sets in $\mathbb{R}^n$: the shape point set $X = \{\vec{x}_i\}_{i=1}^{N_x}(N_x \in \mathbb{N})$ and the model point set $Y = \{\vec{y}_j\}_{j=1}^{N_y}(N_y \in \mathbb{N})$, in order to guarantee the consistency of the shape point set and the model point set in Euclidean distance space, the ICP algorithm is employed to solve the rigid transformation. Whether the two registered point sets are matched, least square (LS) is implemented to measure. The formula can be expressed as follows:

$$\begin{aligned} \min_{R,t,j\in\{1,2,\cdots,N_y\}} \quad & (\textstyle\sum_{i=1}^{N_x} \parallel (\mathbf{R}\vec{x}_i + t) - \vec{y}_j \parallel_2^2) \\ s.t. \quad & \mathbf{R}^{\mathrm{T}}\mathbf{R} = I_n, \det \mathbf{R} = 1 \end{aligned} \tag{1}$$

The procedure of the ICP algorithm includes two steps. At each iterative step, correspondence is set up by finding the model points which are the closest to the shape point set, and then the rigid transformation is solved. The main steps of ICP are summarized as follows:

1) According to the obtained rigid transformation of step $k-1$, the shape point set will be transformed by rotation matrix $\mathbf{R}_{k-1}$ and translation vector t_{k-1}. After transformation, correspondence between two point sets is established as follows:

$$\begin{aligned} c_k(i) = \underset{j\in\{1,2,\cdots,N_y\}}{\arg\min} (\parallel (\mathbf{R}_{k-1}\vec{x}_i + t_{k-1}) - \vec{y}_j \parallel_2^2), \\ i = 1, 2, \cdots, N_x \end{aligned} \tag{2}$$

2) For the shape point set $\{\vec{x}_i\}_{i=1}^{N_x}$ and the model point set $\{\vec{y}_{c_k(i)}\}_{i=1}^{N_x}$ which conform to the correspondence of step 1), we need to solve new rigid transformation between them as follows :

$$(\mathbf{R}_k, t_k) = \underset{\widetilde{\mathbf{R}}_k^{\mathrm{T}}\widetilde{\mathbf{R}}_k = I_n, \det\widetilde{\mathbf{R}}=1, \widetilde{t}_k}{\arg\min} (\sum_{i=1}^{N_x} \parallel (\widetilde{\mathbf{R}}_k\vec{x}_i + \widetilde{t}_k - \vec{y}_{c_k(i)}) \parallel_2^2) \tag{3}$$

Step 1) and 2) are repeated until the iteration is convergent. As the method is a local convergent method, initial values of rotation matrix and translation vector are important. Good initial values not only guarantee the algorithm converges to the global minimum point, but also will greatly improve the efficiency of computation. The selection of initial values also has a lot of methods [1], and it will not be concretely analyzed.

III. PROBABILITY ITERATIVE CLOSEST POINT ALGORITHM

A. Problem Statement

The traditional ICP algorithm can achieve the rigid registration with good accuracy and fast speed, but it fails to register two point sets with noise named noisy point set. The noise got by the sensor or produced by the image processing is called shape noise. Fig. 1 presents a registration result of 2D noisy point sets.

In Fig. 1, the red points show one shape without noise, while the blue points show the noisy point set. Through the ICP registration result, it is obvious to see that the noisy points on the edge will influence the result of registration significantly. Due to the interference of noise, the shape boundaries after registration will not completely aligned, so Gaussian probability model is introduced. The probabilities of points between shape point set and model point set are real-time calculated at each iteration. On the whole the distances of noisy points are far, so probabilities calculated by Gaussian model are small. In this way, noisy points contribute little to the registration result which meets our expectation.

The registration of two point sets with noise is considered as a probability density estimation problem. According to law of total probability, the probability formula of shape point set is as follows:

$$p(\vec{x}) = \sum_{j=1}^{N_y} P(\vec{y}) p(\vec{x}|\vec{y}) \tag{4}$$

For shape point set and model point set satisfying the Gaussian probability model, the following formula is used to show their relationship:

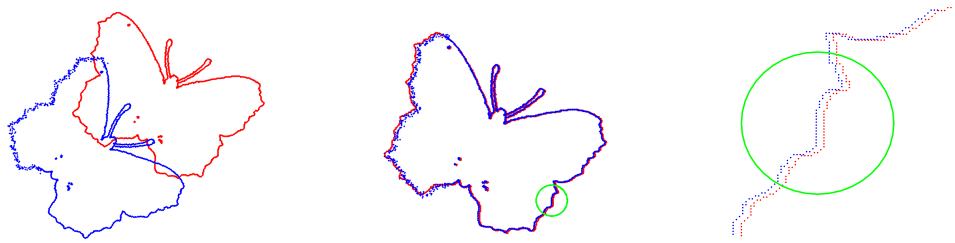
$$p(\vec{x}|\vec{y}) = \frac{1}{(2\pi\sigma^2)^{\frac{n}{2}}} exp^{-\frac{\|T(\vec{x})-\vec{y}\|_2^2}{2\sigma^2}} \tag{5}$$

where n is the dimension of point sets, σ^2 is the variance of the Gaussian probability model and $\parallel T(\vec{x}) - \vec{y} \parallel_2^2$ shows the distance between the transformed shape point set and the model point set.

According to (5), it can be seen that probability formula is Gaussian distribution with zero mean and σ^2 variance. Because $\parallel T(\vec{x}) - \vec{y} \parallel_2^2$ is part of the molecule of the index and is negative, shorter the distance of the transformed shape point set and the model point set is, greater the Gaussian probability and vice versa. The distances of noisy points are far as a whole, so the weights are small. Thus it can gradually reduce the influence of noise, and achieves satisfactory registration result.

Without any other condition, in order to make the model simple, we assume that the model point set conforms to the uniform distribution, hence:

$$p(\vec{y}) = \frac{1}{N_y} \tag{6}$$



(a) (b) (c)

Fig. 1. Point sets of different shapes. (red: point set is without noise, blue: point set is the noisy point set) (a) 2D Butterfly. (b) Registration result of ICP. (c) Amplification result of the green area.

Because Gaussian probability model is introduced, the variance which changes can be seemed as a parameter considering the influence on accuracy and speed for maximum-likelihood (ML) estimation. Objective function of the likelihood function is as follows:

$$F(\vec{x},\mathbf{R},t,\sigma^2) = -\sum_{i=1}^{N_x}\log p(\vec{x}_i|\mathbf{R},t,\sigma^2)$$
$$= -\sum_{i=1}^{N_x} log\sum_{j=1}^{N_y} P(\vec{y}_j)p(\vec{x}_i|\vec{y}_j,\mathbf{R},t,\sigma^2) \quad (7)$$

B. Probability Iterative Closest Point Algorithm

In this paper, we study the rigid registration for noisy points. Similar to the EM-ICP algorithm [11], we also introduce the EM principle to the ICP algorithm to solve the problem (7). EM consists of two steps: E-Step and M-Step. Hence the proposed algorithm is also composed of two parts. During the E-Step, the one-to-one correspondence is established between the shape point set and the model point set. M-Step is to solve the latest rigid transformation based on E-Step using singular value decomposition (SVD) method, and then the Gaussian probability is calculated according to the distance and variance between two point sets. Establishing the one-to-one correspondence between two point sets can greatly reduce the computing time and achieve high efficiency. Moreover, the variance of Gaussian probability model is updated from large to small step by step, which can make the registration from global to local to prevent the proposed algorithm from being trapped into the local minimum.

EM algorithm is used to solve (7) including E-Step and M-Step:

E-Step: Based on the rigid transformation of $(k-1)^{th}$ step, the correspondence of kth step is established adopting one-to-one correspondence which is for each shape point to find the nearest model point:

$$\min\sum_{i=1}^{N_x} \parallel \mathbf{R}_{k-1}\vec{x}_i + t_{k-1} - \vec{y}_{c_k(i)} \parallel_2^2 \quad (8)$$

Delaunay triangulation and k-d tree are common closest point search strategies, and in this paper the former is employed.

M-Step: The posterior probability got by (15) is employed as the probability, and the initial variance is given a big value, so it can be assumed that the initial distribution is uniform distribution. As correspondence and new corresponding Gaussian probability are known, (7) is used to establish a new objective function. By maximizing the objective function, the new rigid transformation is solved including rotation matrix $\mathbf{R}_k$, translation vector t_k and variance σ_k^2. The new objective function is established as follows:

$$M(\tilde{\mathbf{R}}_k,\tilde{t}_k,\tilde{\sigma}_k^2) = \sum_{i=1}^{N_x}\sum_{j=1}^{N_y} p(\vec{y}_j|\vec{x}_i,\mathbf{R}_{k-1},t_{k-1},\sigma_{k-1}^2)$$
$$\times log(P(\vec{y}_j)p(\vec{x}_i|\vec{y}_j,\tilde{\mathbf{R}}_k,\tilde{t}_k,\tilde{\sigma}_k^2)) \quad (9)$$

For the one-to-one correspondence is adopted, the probability is not zero only when $j = c_k(i)$, so (9) can be simplified as follows:

$$M(\tilde{\mathbf{R}}_k,\tilde{t}_k,\tilde{\sigma}_k^2) = \sum_{i=1}^{N_x} p(\vec{y}_{c_k(i)}|\vec{x}_i,\mathbf{R}_{k-1},t_{k-1},\sigma_{k-1}^2)$$
$$\times log(P(\vec{y}_{c_k(i)})p(\vec{x}_i|\vec{y}_{c_k(i)},\tilde{\mathbf{R}}_k,\tilde{t}_k,\tilde{\sigma}_k^2)) \quad (10)$$

Let $p_i^{k-1} = p(\vec{y}_{c_k(i)}|\vec{x}_i,\mathbf{R}_{k-1},t_{k-1},\sigma_{k-1}^2)$, and the new target function is got by simplifying (10) using (5), (6) and (15):

$$M(\tilde{\mathbf{R}}_k,\tilde{t}_k,\tilde{\sigma}_k^2) = -\sum_{i=1}^{N_x} p_i^{k-1}\times\frac{\parallel \tilde{\mathbf{R}}_k\vec{x}_i+\tilde{t}_k-\vec{y}_{c_k(i)} \parallel}{2\tilde{\sigma}_k^2}$$
$$-\sum_{i=1}^{N_x} p_i^{k-1}\times\log(N_y(2\pi\tilde{\sigma}_k^2)^{\frac{n}{2}}) \quad (11)$$

Because of only the first part related to the rotation matrix and translation vector, maximizing above formula is equivalent to maximizing $-p_i^{k-1}\times\frac{\|\tilde{\mathbf{R}}_k\vec{x}_i+\tilde{t}_k-\vec{y}_{c_k(i)}\|}{2\tilde{\sigma}_k^2}$ with respect to the rotation matrix and translation vector, and $\frac{1}{N_y}$ is a uniform distribution, so the goal is to solve the following function:

$$(\mathbf{R}_k,t_k) = \underset{\substack{\tilde{\mathbf{R}}_k^T\tilde{\mathbf{R}}_k=I_n,\\ \det\tilde{\mathbf{R}}=1,\tilde{t}_k}}{\arg\min}\sum_{i=1}^{N_x} p_i^{k-1}\parallel(\tilde{\mathbf{R}}_k\vec{x}_i+\tilde{t}_k)-\vec{y}_{c_k(i)}\parallel_2^2 \quad (12)$$

The form of (12) is similar to that of the classical ICP algorithm just more than a probability p_i^{k-1}, and can be solved by SVD. For the probability is introduced in this paper, we define the root mean square (RMS) error as

$(\sum_{i=1}^{N_x} p_i \parallel (\mathbf{R}_k\vec{x}_i+t_k)-\vec{y}_{c_k(i)} \parallel_2^2)^{0.5}$. The EM mean square variance adopts solving the derivation of $M(\mathbf{R}_k, t_k, \tilde{\sigma}_k^2)$ based on the variance and the derivation is 0 corresponding to the variance desired. The EM mean square variance is obtained as follows:

$$\sigma_{EM}^2 = \sum_{i=1}^{N_x} p_i^{k-1} \times \frac{\parallel (\mathbf{R}_k\vec{x}_i + t_k) - \vec{y}_{c_k(i)} \parallel_2^2}{n} \tag{13}$$

The EM mean square variance σ_{EM}^2 obtained from (13) is actually the registration error of kth step.

After getting EM mean square variance, (14) is used to update the variance of the Gaussian probability

$$\sigma_k^2 = \begin{cases} \sigma_{k-1}^2/\lambda & \sigma_{k-1}^2/\lambda > \sigma_{EM}^2 \\ \sigma_{EM}^2 & \sigma_{k-1}^2/\lambda < \sigma_{EM}^2 \end{cases} \tag{14}$$

λ is called annealing coefficient. When λ is 1, it is a special case which is not updating the variance, so it is standard ICP algorithm. When λ is big, it can improve the convergence speed, but it's also easy to fall into local minimum. Therefore, the value of λ is range from 1 to 2 which is selected by experiments. To make full use of information of all points preventing the algorithm from being trapped in local minimum, the updating of variance adopts the idea of from global to local. In the beginning the variance is given a big value, so all the points are approximate to uniform distribution, and σ_{k-1}^2/λ is used to update the variance of the Gaussian probability model from large to small step by step. When the two images are roughly registered, the PICP distinguishes between noisy points and valid points according to the distance and variance between two point sets. Hence the EM mean square variance which is actually the registration error of last step is employed to precisely update the variance of the Gaussian probability model. The distances of noisy points are far as a whole, so the weights are small. Thus it can gradually reduce the influence of noise, and achieves satisfactory registration result.

Using $\mathbf{R}_k$, t_k and σ_k^2 of this step, the Gaussian probabilities are updated between corresponding points and the weights are calculated:

$$p(\vec{y}_{c_k(i)}|\vec{x}_i,\mathbf{R}_k, t_k, \sigma_k^2) = \frac{1}{(2\pi\sigma_k^2)^{\frac{n}{2}} P} exp - \frac{\parallel \mathbf{R}_k\vec{x}_i + t_k - \vec{y}_{c_k(i)} \parallel_2^2}{2\sigma_k^2} \tag{15}$$

where

$$P = \sum_{i=1}^{N_x} \frac{1}{(2\pi\sigma_k^2)^{\frac{n}{2}}} exp - \frac{\parallel \mathbf{R}_k\vec{x}_i + t_k - \vec{y}_{c_k(i)} \parallel_2^2}{2\sigma_k^2}$$

is used to normalize the probability.

Establishing correspondence between all the points reduces the efficiency, so the one-to-one correspondence is used to find the model point nearest to the shape point. In order to avoid the algorithm being trapped into local minimum, the variance of the Gaussian probability updates adopting the idea of from global to local. The core of this algorithm is setting up the Gaussian probability model and updating it automatically, so the algorithm is named probability-ICP (PICP), which is summarized in TABLE I.

TABLE I THE PROCESS OF THE PROPOSED PICP ALGORITHM

Algorithm 1: The proposed PICP algorithm.

1. Give the initial transformation $\mathbf{R}_0 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $t_0 = (0,0)^T$, set the value of $\lambda \in (1,2]$, and set the initial probability $p_i^0 = \frac{1}{N_x}, i = 1, 2, \cdots, N_x$.
2. Set $X = \mathbf{R}_0 X + t_0$ and the iterative index $k = 0$.
3. **REPEAT**
4. $k = k + 1$
5. Set up the correspondence $c_k(i), i \in \{1, 2, \cdots, N_x\}$.
6. Solve the rotation matrix $\mathbf{R}_k$ and translation vector t_k by SVD.
7. Compute the variance between two point set by EM method $\sigma_{EM}^2 = \sum_{i=1}^{N_x} p_i^{k-1} \times \frac{\|(\mathbf{R}_k\vec{x}_i+t_k)-\vec{y}_{c_k(i)}\|_2^2}{n}$
8. Update the variance of the Gaussian probability $\sigma_k^2 = \begin{cases} \sigma_{k-1}^2/\lambda & \sigma_{k-1}^2/\lambda > \sigma_{EM}^2 \\ \sigma_{EM}^2 & \sigma_{k-1}^2/\lambda < \sigma_{EM}^2 \end{cases}$
9. Update the Gaussian probability model by the rotation matrix $\mathbf{R}_k$, translation vector t_k and variance σ_k^2 $p_i^k = \frac{1}{(2\pi\sigma_k^2)^{\frac{n}{2}} P} exp - \frac{\|\mathbf{R}_k\vec{x}_i+t_k-\vec{y}_{c_k(i)}\|_2^2}{2\sigma_k^2}$
10. **UNTIL** $RMS_k - RMS_{k-1} < \varepsilon$ or $k > Step_{max}$.

At the end of each iteration, the algorithm needs to judge if it meets exiting conditions. If the difference between the RMS_k and RMS_{k-1} is smaller than a small number ε or the number of iteration k reaches the given maximum $Step_{max}$, the algorithm has reached the optimal minimum, so the program needs to terminate.

In the PICP all correspondence is improved as the one-to-one point correspondence. To prevent the proposed algorithm from being trapped into the local minimum, the variance of Gaussian probability model is updated from large to small step by step, which can make the registration from global to local. If the PICP algorithm adopts EM mean square variance to update the variance of Gaussian probability model from the start, the registration process is very difficult to make full use of information of all points, and is easy to fall into the local minimum. So we first reduce variance through the constant proportion which is just the annealing coefficient λ until the two point sets are roughly matched well. Then EM mean square variance further accurately updates the variance of Gaussian probability model which not only ensures high precision, but also has very good robustness.

IV. EXPERIMENTAL RESULTS

In this section the PICP algorithm is compared with the ICP and EM-ICP algorithm. Experiments are conducted on the part B of CE-Shape-1 [12] and in the position estimation in intelligent vehicle system platform of Xi'an Jiaotong University.

A. Standard Database

TABLE II shows RMS of Apple, Cup and Children. It can be seen that RMS of PICP is the minimum, because when the two shapes are close enough, the weights of noisy points will be smaller and the weights of real points will be

TABLE II RMS COMPARISON OF 2D POINT SETS

Point sets	ICP	EM-ICP	PICP
Apple	7.3768	7.2094	0.0035
Cup	4.3299	3.9069	0.0047
Children	1.4703	1.4144	0.0057

bigger greatly reducing influence of noise on the result of registration.

Fig. 2 is the result using different registration algorithms for the data Apple from part B of CE-Shape-1, and it can be seen that PICP registration algorithm performs best and it deals with the noise effectively. Fig. 3 shows the convergence of three kinds of registration algorithms. It can be seen EM-ICP algorithm converges slowly and PICP algorithm converges fast, because PICP algorithm compared with EM-ICP algorithm is improved from all correspondence to one-to-one correspondence. In the early stage, the convergence of ICP and PICP is almost the same, because in the beginning Gaussian probability model given big variance is amount to uniform distribution. Noisy points and real points have roughly equal probabilities, so PICP algorithm degenerates into ICP algorithm. When two shapes are roughly registered, the value got by EM is used to update Gaussian probability model and improves the convergence precision. PICP gives the best registration, and the reason is when the two image point sets are close enough, the automatically updated Gaussian probability decreases the probabilities of the noisy points and increases the probabilities of the real points. Therefore, the influence of noise on the result of registration is effectively suppressed and the result is more accurate.

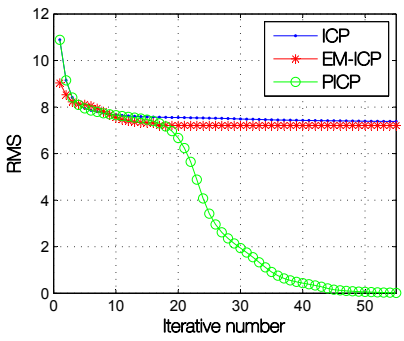


Fig. 3. Various registration convergence of Apple with noise.

B. Application in the Position Estimation

In this part, the PICP algorithm is applied to the position estimation in intelligent vehicle system verification platform of Xi'an Jiaotong University.

The Global Position System (GPS) and the local positioning system consisting of an optical fiber gyroscope and a speed sensor are commonly used. Positioning by GPS does not produce cumulative error, but in the environment of many trees and buildings, signals of satellites are vulnerable to interference which leads to noise, even missing some data. Local positioning is accurate for a short time hardly affected by the external environment, but it has cumulative error. In order to combine with the advantages of GPS global positioning and local positioning, GPS global routes and local routes will be registered by PICP to get revised global routes which are real-time, accurate and smooth.

Fig. 4 shows the GPS global position and the local position. In Fig. 4(b)-(c), A and B of GPS are amplified where the data has much noise and is very not smooth for the poor signal obscured by trees.

Since the local position has cumulative error, it is impossible to register all the data once, thus the step-by-step method for registration is employed. At every step, the transformation can be got by registering the local data to global data of last 50 meters by the PICP algorithm, and then the transformation is used on the new local position to predict the latest global position. All the results are the transformed local points which are smooth.

Fig. 5 is the registration result by the PICP algorithm. Through the amplification of A and B, it can be seen the route becomes smooth and the wrong points for trees sheltering are rectified which is very valuable in the position estimation. In the first stage of registration, the weight of all points are the same, but when two graphics are close enough, PICP algorithm will give weight according to the distance between point and point. For effective points, distances are short, so they get heavy weights. For noisy points, distances are long, so they get small weights. It effectively reduces the impact of GPS noise on registration results making the final route smooth and accurate.

V. CONCLUSION

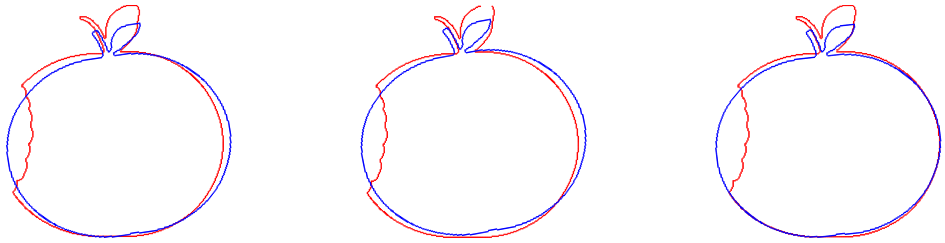
This paper addressed the problem of registering point sets with shape noise by the PICP algorithm based on EM principle. Compared with all correspondence of EM-ICP, PICP algorithm adopts the one-to-one correspondence which greatly improves calculation efficiency. Updating the variance of Gaussian probability model employs the idea of from global to local. By introducing annealing coefficient, PICP algorithm will not be trapped in local minima. When two shapes are close enough, the PICP distinguishes between noisy points and valid points according to the distance and variance between two point sets, so that it reduces the influence of noise on registration results. Then, some experiments of 2D registration result are presented to illustrate that the PICP performance is better than the ICP and EM-ICP. Finally the PICP algorithm is applied to the position estimation which effectively eliminates noise and gets smooth and precise results.

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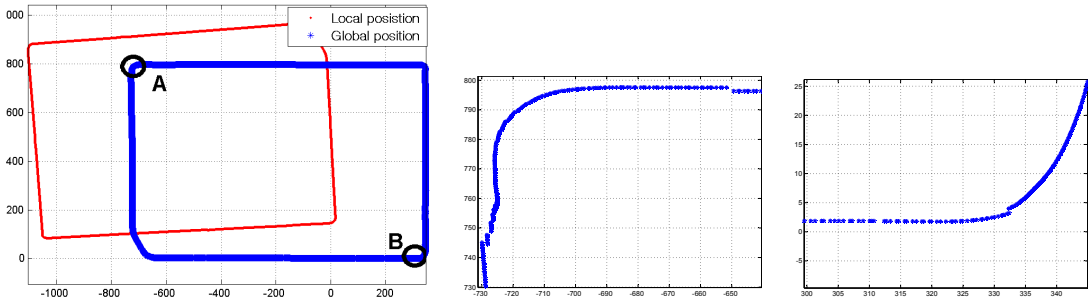
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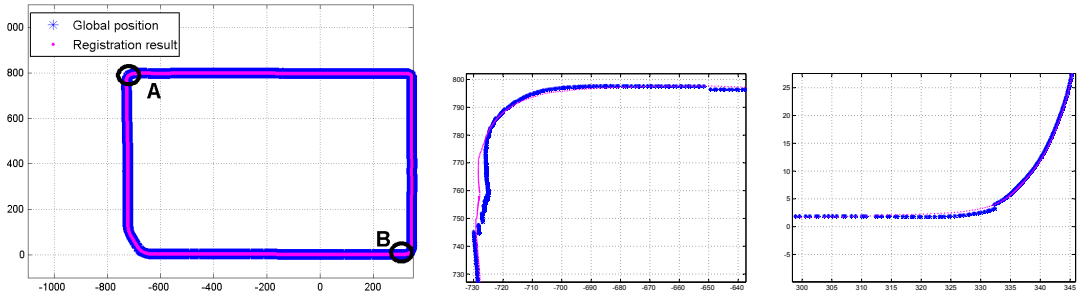
(a) (b) (c)

Fig. 2. Registration result of image point sets Apple. (a) ICP registration result. (b) EM-ICP registration result. (c) PICP registration result.



(a) (b) (c)

Fig. 4. (a) The GPS global position and the local position. (b) Amplification result of A. (c) Amplification result of B.



(a) (b) (c)

Fig. 5. (a) The PICP registration result of GPS global position and the local position. (b) Amplification result of A. (c) Amplification result of B.

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